

Analysis of Linear Programming

Part 3f: Sensitivity and Post-Optimality Analysis

IE 311 – Operations Research I

- So far we have learned how to model LPs and solve them.
- We now discuss how to **interpret** the results and understand how they would change due to changes in the problem data.
- In particular, we answer two types of questions:
- **Sensitivity:** how much can we change the data of the problem (objective vector, right-hand side, or constraint matrix) so that the optimal basis does **not** change?
- **Post-Optimality:** if the optimal basis **does** need to change, how can we re-optimize to find the new optimal solution *using the previous optimal basis*?

Recall: Matrix Form of the Simplex Tableau (Summary)

- Consider an LP in standard form:
$$\begin{aligned} \max \quad & x_0 = cx \\ \text{s.t.} \quad & Ax = b \\ & x \geq 0. \end{aligned}$$
- Assume that the **optimal tableau** is available (BV is the optimal basis):

Row	BV		RHS
0	x_0	$x_0 + (c_{\text{BV}}B^{-1}N - c_{\text{NBV}})x_{\text{NBV}}$	$= c_{\text{BV}}B^{-1}b$
$1, \dots, m$	x_{BV}	$x_{\text{BV}} + (B^{-1}N)x_{\text{NBV}}$	$= B^{-1}b$

- Recall that we must have $B^{-1}b \geq 0$ and $\bar{c} := c_{\text{BV}}B^{-1}N - c_{\text{NBV}} \geq 0$.
- Notation: e_j (and e_j^T) is the j -th unit column (and row)-vector.
- Notation: (j) is the row of the basic variable x_j .

- **Question:** how much can we change the data of the problem (objective vector, right-hand side, or constraint matrix) so that the optimal basis does not change?
- We consider the following cases and derive conditions so that the optimal basis stays the same:
 - changing the objective coefficient of a nonbasic / basic variable;
 - changing the right-hand side of a constraint;
 - changing the column of a nonbasic / basic variable.
- These analyses tell us how **sensitive** the optimal solution is to data perturbations.

- Suppose that we change the objective coefficient of a nonbasic variable x_j by Δ , that is, $c_j \rightarrow c_j + \Delta$.
- Question:** Find conditions for Δ so that the current basis is still optimal.
- Answer:** Changing the objective coefficient does not affect (primal) feasibility, but it might affect dual feasibility (i.e. optimality).
 - Recall the reduced cost $\bar{c}_j := c_{BV}B^{-1}A^j - c_j \geq 0$.
 - New reduced cost becomes $c_{BV}B^{-1}A^j - (c_j + \Delta) = \bar{c}_j - \Delta$.
 - The current basis is still optimal if $\bar{c}_j - \Delta \geq 0$.

- Suppose that we change the objective coefficient of a basic variable x_j by Δ , that is, $c_j \rightarrow c_j + \Delta$ (or $c_{BV} \rightarrow c_{BV} + \Delta e_{(j)}^T$).
- **Question:** Find conditions for Δ so that the current basis is still optimal.
- **Answer:** Changing the objective coefficient does not affect (primal) feasibility, but it might affect dual feasibility (i.e. optimality).
 - Recall the reduced cost of nonbasic variables $\bar{c} := c_{BV}B^{-1}N - c_{NBV} \geq 0$.
 - New reduced cost becomes $(c_{BV} + \Delta e_{(j)}^T)B^{-1}N - c_{NBV} = \bar{c} + \Delta e_{(j)}^T B^{-1}N$.
 - The current basis is still optimal if $\bar{c} + \Delta e_{(j)}^T B^{-1}N \geq 0$.

- Suppose that we change the right-hand side of constraint i by Δ , that is, $b_i \rightarrow b_i + \Delta$ (or $b \rightarrow b + \Delta e_i$).
- **Question:** Find conditions for Δ so that the current basis is still optimal.
- **Answer:** Changing the right-hand side of a constraint does not affect dual feasibility, but it might affect (primal) feasibility.
 - Recall the RHS of the tableau (value of basic variables) $x_{BV}^* = B^{-1}b \geq 0$.
 - New RHS of the tableau becomes $B^{-1}(b + \Delta e_i) = x_{BV}^* + \Delta B^{-1}e_i$.
 - The current basis is still optimal if $x_{BV}^* + \Delta B^{-1}e_i \geq 0$.

- Suppose that we change the column of a nonbasic variable x_j by the vector Δ , that is, $A^j \rightarrow A^j + \Delta$.
- Question:** Find conditions for Δ so that the current basis is still optimal.
- Answer:** Changing column of a nonbasic does not affect (primal) feasibility, but it might affect dual feasibility (i.e. optimality).
 - Recall the reduced cost $\bar{c}_j := c_{BV}B^{-1}A^j - c_j \geq 0$.
 - New reduced cost becomes $c_{BV}B^{-1}(A^j + \Delta) - c_j = \bar{c}_j + c_{BV}B^{-1}\Delta$.
 - The current basis is still optimal if $\bar{c}_j + c_{BV}B^{-1}\Delta \geq 0$.

- Suppose that we change the column of a basic variable x_j by the vector Δ , that is, $A^j \rightarrow A^j + \Delta$.
- **Question:** Find conditions for Δ so that the current basis is still optimal.
- **Answer:** Changing the column of a basic might affect both (primal) feasibility and dual feasibility (i.e. optimality) since the basis matrix changes.
 - We will not analyze this case in the scope of this course.

- Recall the Giapetto LP:

$$x_0^* = \max 3x_1 + 2x_2$$

$$\text{s.t. } 2x_1 + x_2 \leq 100$$

$$x_1 + x_2 \leq 80$$

- Here is the optimal tableau: $x_1, x_2 \geq 0$.

Row	BV				RHS
0	x_0	x_0	$+x_3$	$+x_4$	$= 180$
1	x_1	x_1	$+x_3$	$-x_4$	$= 20$
2	x_2	x_2	$-x_3$	$+2x_4$	$= 60$

Sensitivity Analysis: Giapetto Example (I)

Row	BV				RHS
0	x_0	x_0	$+x_3$	$+x_4$	$= 180$
1	x_1	x_1	$+x_3$	$-x_4$	$= 20$
2	x_2	x_2	$-x_3$	$+2x_4$	$= 60$

Find the range for the profit of x_1 to change in such a way that the current basis remains optimal.

- Suppose that we change c_1 by Δ , that is, $c_1 \rightarrow c_1 + \Delta$.
- Recall the reduced cost of nonbasic variables $\bar{c} := c_{BV}B^{-1}N - c_{NBV} \geq 0$.
- New reduced cost becomes $(c_{BV} + \Delta e_1^T)B^{-1}N - c_{NBV} = \bar{c} + \Delta e_1^T B^{-1}N$.
- The current basis is still optimal if $\bar{c} + \Delta e_1^T B^{-1}N \geq 0$.
- So, we must have $[1 \quad 1] + \Delta[1 \quad 0] \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix} \geq [0 \quad 0]$, or equivalently, $-1 \leq \Delta \leq 1$.

Sensitivity Analysis: Giapetto Example (II)

Row	BV				RHS
0	x_0	x_0	$+x_3$	$+x_4$	$= 180$
1	x_1	x_1	$+x_3$	$-x_4$	$= 20$
2	x_2	x_2	$-x_3$	$+2x_4$	$= 60$

- Find the range for the profit of x_2 to change in such a way that the current basis remains optimal.
 - Suppose that we change c_2 by Δ , that is, $c_2 \rightarrow c_2 + \Delta$.
 - Recall the reduced cost of nonbasic variables $\bar{c} := c_{BV}B^{-1}N - c_{NBV} \geq 0$.
 - New reduced cost becomes $(c_{BV} + \Delta e_2^T)B^{-1}N - c_{NBV} = \bar{c} + \Delta e_2^T B^{-1}N$.
 - The current basis is still optimal if $\bar{c} + \Delta e_2^T B^{-1}N \geq 0$.
 - So, we must have $[1 \quad 1] + \Delta[0 \quad 1] \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix} \geq [0 \quad 0]$, or equivalently, $-\frac{1}{2} \leq \Delta \leq 1$

Sensitivity Analysis: Giapetto Example (III)

Row	BV				RHS
0	x_0	x_0	$+x_3$	$+x_4$	$= 180$
1	x_1	x_1	$+x_3$	$-x_4$	$= 20$
2	x_2	x_2	$-x_3$	$+2x_4$	$= 60$

Find the range for the right-hand side of the first constraint to change in such a way that the current basis remains optimal.

- Suppose that we change b_1 by Δ , that is, $b_1 \rightarrow b_1 + \Delta$.
- Recall the RHS of the tableau (value of basic variables) $x_{BV}^* = B^{-1}b \geq 0$.
- New RHS of the tableau becomes $B^{-1}(b + \Delta e_1) = x_{BV}^* + \Delta B^{-1}e_1$.
- The current basis is still optimal if $x_{BV}^* + \Delta B^{-1}e_1 \geq 0$.
- So, we must have $\begin{bmatrix} 20 \\ 60 \end{bmatrix} + \Delta \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} \geq \begin{bmatrix} 0 \\ 0 \end{bmatrix}$, or equivalently, $-20 \leq \Delta \leq 60$.

Sensitivity Analysis: Giapetto Example (IV)

Row	BV				RHS
0	x_0	x_0	$+x_3$	$+x_4$	$= 180$
1	x_1	x_1	$+x_3$	$-x_4$	$= 20$
2	x_2	x_2	$-x_3$	$+2x_4$	$= 60$

Find the range for the right-hand side of the second constraint to change in such a way that the current basis remains optimal.

- Suppose that we change b_2 by Δ , that is, $b_2 \rightarrow b_2 + \Delta$.
- Recall the RHS of the tableau (value of basic variables) $x_{BV}^* = B^{-1}b \geq 0$.
- New RHS of the tableau becomes $B^{-1}(b + \Delta e_2) = x_{BV}^* + \Delta B^{-1}e_2$.
- The current basis is still optimal if $x_{BV}^* + \Delta B^{-1}e_2 \geq 0$.
- So, we must have $\begin{bmatrix} 20 \\ 60 \end{bmatrix} + \Delta \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} \geq \begin{bmatrix} 0 \\ 0 \end{bmatrix}$, or equivalently, $-30 \leq \Delta \leq 20$.

- Recall the Dakota Furniture Company LP:

$$x_0^* = \max 60x_1 + 30x_2 + 20x_3$$

$$\text{s.t. } 8x_1 + 6.5x_2 + 1.5x_3 \leq 48$$

$$4x_1 + 2.5x_2 + 1.5x_3 \leq 20$$

$$2x_1 + 1.5x_2 + 0.5x_3 \leq 8$$

- Obtain the optimal tableau using $x_1, x_2, x_3 \geq 0$.

SimplexTableauWithBV.py.

- Carry out the tasks in HandoutSensitivity (go to Sucourse):**

- Find the range for the profit of each variable to change in such a way that the current basis remains optimal.
- Find the range for the right-hand side of each constraint to change in such a way that the current basis remains optimal.

- **Question:** if the optimal basis needs to change, how can we re-optimize to find the new optimal solution *using the previous optimal basis*?
- We consider the following cases (assuming the optimal basis must change):
- changing the objective coefficient of a nonbasic / basic variable;
- changing the right-hand side of a constraint;
- changing the column of a nonbasic variable;
- adding a new variable (column);
- adding a new constraint (row).
- These analyses tell us how to obtain a new optimal solution after data perturbations **without solving the problem from scratch.**

- Suppose that we change the objective coefficient of a nonbasic variable x_j by Δ , (that is, $c_j \rightarrow c_j + \Delta$) such that

$$c_{BV}B^{-1}A^j - (c_j + \Delta) = \bar{c}_j - \Delta \not\geq 0.$$

- **Question:** Re-optimize to find the new optimal solution.
- **Answer:** Changing the objective coefficient does not affect (primal) feasibility, but it might affect dual feasibility (i.e. optimality).
 - Therefore, we can continue with primal simplex method (choose x_j as the entering variable) after updating the reduced cost.

- Suppose that we change the objective coefficient of a basic variable x_j by Δ , (that is, $c_j \rightarrow c_j + \Delta$ or $c_{BV} \rightarrow c_{BV} + \Delta e_{(j)}^T$) such that

$$(c_{BV} + \Delta e_{(j)}^T)B^{-1}N - c_{NBV} = \bar{c} + \Delta e_{(j)}^T B^{-1}N \not\geq 0.$$

- Question:** Re-optimize to find the new optimal solution.
- Answer:** Changing the objective coefficient does not affect (primal) feasibility, but it might affect dual feasibility (i.e. optimality).
 - Therefore, we can continue with primal simplex method (choose the nonbasic variable with the most negative reduced cost as the entering variable) after updating the reduced cost.

- Suppose that we change the right-hand side of constraint i by Δ , (that is, $b_i \rightarrow b_i + \Delta$ or $b \rightarrow b + \Delta e_i$) such that

$$B^{-1}(b + \Delta e_i) = x_{BV}^* + \Delta B^{-1}e_i \not\geq 0.$$

- Question:** Re-optimize to find the new optimal solution.
- Answer:** Changing the right-hand side of a constraint does not affect dual feasibility, but it might affect (primal) feasibility.
 - Therefore, we can continue with dual simplex method (choose the basic variable with the most negative RHS value as the leaving variable) after updating the RHS.

- Suppose that we change the column of a nonbasic variable x_j by the vector Δ , (that is, $A^j \rightarrow A^j + \Delta$) such that

$$c_{BV}B^{-1}(A^j + \Delta) - c_j = \bar{c}_j + c_{BV}B^{-1}\Delta \not\geq 0.$$

- **Question:** Re-optimize to find the new optimal solution.
- **Answer:** Changing the column of a nonbasic does not affect (primal) feasibility, but it might affect dual feasibility (i.e. optimality).
 - Therefore, we can continue with primal simplex method (choose x_j as the entering variable) after updating the reduced cost and the column of the variable x_j .

- Suppose that we add a new nonnegative variable x_j with the *column* A^j and the objective coefficient c_j where $\bar{c}_j := c_{BV}B^{-1}A^j - c_j$.
- **Question:** Re-optimize to find the new optimal solution.
- **Answer:** Adding a new nonnegative variable does not affect (primal) feasibility, but it might affect dual feasibility (i.e. optimality).
 - If $\bar{c}_j \geq 0$, then the new variable x_j is nonbasic and the previous basis remains optimal.
 - Otherwise, we can continue with primal simplex method (choose x_j as the entering variable) after updating the reduced cost and the column of the variable x_j .

- Suppose that we add a new constraint.
- **Question:** Re-optimize to find the new optimal solution.
- **Answer:** Adding a constraint does not affect dual feasibility, but it might affect (primal) feasibility.
 - If the current optimal solution satisfies the new constraint, then it remains optimal.
 - Otherwise, we add the new constraint (in standard form with the addition of slack/surplus variables if needed) to the tableau and zero-out the coefficients of the basic variables in the newly added row. Finally, we can continue with dual simplex method.

- Recall the Giapetto LP:

$$x_0^* = \max 3x_1 + 2x_2$$

$$\text{s.t. } 2x_1 + x_2 \leq 100$$

$$x_1 + x_2 \leq 80$$

- Here is the optimal tableau: $x_1, x_2 \geq 0$.

Row	BV				RHS
0	x_0	x_0	$+x_3$	$+x_4$	$= 180$
1	x_1	x_1	$+x_3$	$-x_4$	$= 20$
2	x_2	x_2	$-x_3$	$+2x_4$	$= 60$

Row	BV				RHS
0	x_0	x_0	$+x_3$	$+x_4$	$= 180$
1	x_1	x_1	$+x_3$	$-x_4$	$= 20$
2	x_2	x_2	$-x_3$	$+2x_4$	$= 60$

- Suppose that the profit of x_1 is **increased by 2 units**
(recall from the sensitivity analysis that the current basis remains optimal if the amount of change Δ satisfies $-1 \leq \Delta \leq 1$).
- We need to update the reduced cost as $(c_{BV} + \Delta e_1^T)B^{-1}N - c_{NBV} = \bar{c} + \Delta e_1^T B^{-1}N = [1 \quad 1] + 2[1 \quad 0] \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix} = [3 \quad -1]$.
- We need to update the objective value as $(c_{BV} + \Delta e_1^T)B^{-1}b = 220$.

Modified Tableau:

Row	BV				RHS
0	x_0	x_0	$+3x_3$	$-x_4$	$= 220$
1	x_1	x_1	$+x_3$	$-x_4$	$= 20$
2	x_2	x_2	$-x_3$	$+2x_4$	$= 60$

- x_4 enters, x_2 leaves.
- New Tableau (also optimal):

Row	BV				RHS
0	x_0	x_0	$+\frac{1}{2}x_2$	$+\frac{5}{2}x_3$	$= 250$
1	x_1	x_1	$+\frac{1}{2}x_2$	$+\frac{1}{2}x_3$	$= 50$
2	x_4		$\frac{1}{2}x_2$	$-\frac{1}{2}x_3$	$+x_4$ $= 30$

Row	BV				RHS
0	x_0	x_0	$+x_3$	$+x_4$	$= 180$
1	x_1	x_1	$+x_3$	$-x_4$	$= 20$
2	x_2	x_2	$-x_3$	$+2x_4$	$= 60$

- Suppose that the profit of x_2 is **decreased by 1 unit**
(recall from the sensitivity analysis that the current basis remains optimal if the amount of change Δ satisfies $-\frac{1}{2} \leq \Delta \leq 1$).
- We need to update the reduced cost as $(c_{BV} + \Delta e_2^T)B^{-1}N - c_{NBV} = \bar{c} + \Delta e_2^T B^{-1}N = [1 \quad 1] - 1[0 \quad 1] \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix} = [2 \quad -1]$.
- We need to update the objective value as $(c_{BV} + \Delta e_2^T)B^{-1}b = 120$.

Modified Tableau:

Row	BV				RHS
0	x_0	x_0	$+2x_3$	$-x_4$	$= 120$
1	x_1	x_1	$+x_3$	$-x_4$	$= 20$
2	x_2	x_2	$-x_3$	$+2x_4$	$= 60$

- x_4 enters, x_2 leaves.
- New Tableau (also optimal):

Row	BV				RHS
0	x_0	x_0	$+\frac{1}{2}x_2$	$+\frac{3}{2}x_3$	$= 150$
1	x_1	x_1	$+\frac{1}{2}x_2$	$+\frac{1}{2}x_3$	$= 50$
2	x_4		$\frac{1}{2}x_2$	$-\frac{1}{2}x_3$	$+x_4$ $= 30$

Row	BV				RHS
0	x_0	x_0	$+x_3$	$+x_4$	$= 180$
1	x_1	x_1	$+x_3$	$-x_4$	$= 20$
2	x_2	x_2	$-x_3$	$+2x_4$	$= 60$

- Suppose that the RHS of constraint 1 is **increased by 80 units** (recall from the sensitivity analysis that the current basis remains optimal if the amount of change Δ satisfies $-20 \leq \Delta \leq 60$).
- We need to update the RHS as $B^{-1}(b + \Delta e_1) = x_{BV}^* + \Delta B^{-1}e_1 = \begin{bmatrix} 20 \\ 60 \end{bmatrix} + 80 \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 100 \\ -20 \end{bmatrix}$.
- We need to update the objective value as $c_{BV}B^{-1}(b + \Delta e_1) = 260$.

Modified Tableau:

Row	BV					RHS
0	x_0	x_0		$+x_3$	$+x_4$	$= 260$
1	x_1		x_1	$+x_3$	$-x_4$	$= 100$
2	x_2		x_2	$-x_3$	$+2x_4$	$= -20$

- x_2 leaves, x_3 enters.
- New Tableau (also optimal):

Row	BV					RHS
0	x_0	x_0		$+x_2$	$+3x_4$	$= 240$
1	x_1		x_1	$+x_2$	$+x_4$	$= 80$
2	x_3			$-x_2$	$+x_3$	$= 20$

Row	BV				RHS
0	x_0	x_0	$+x_3$	$+x_4$	$= 180$
1	x_1	x_1	$+x_3$	$-x_4$	$= 20$
2	x_2	x_2	$-x_3$	$+2x_4$	$= 60$

- Suppose that the RHS of constraint 2 is **decreased by 40 units** (recall from the sensitivity analysis that the current basis remains optimal if the amount of change Δ satisfies $-30 \leq \Delta \leq 20$).
- We need to update the RHS as $B^{-1}(b + \Delta e_2) = x_{BV}^* + \Delta B^{-1}e_2 = \begin{bmatrix} 20 \\ 60 \end{bmatrix} - 40 \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 60 \\ -20 \end{bmatrix}$.
- We need to update the objective value as $c_{BV}B^{-1}(b + \Delta e_2) = 140$.

Modified Tableau:

Row	BV				RHS
0	x_0	x_0	$+x_3$	$+x_4$	$= 140$
1	x_1	x_1	$+x_3$	$-x_4$	$= 60$
2	x_2	x_2	$-x_3$	$+2x_4$	$= -20$

- x_2 leaves, x_3 enters.
- New Tableau (also optimal):

Row	BV				RHS
0	x_0	x_0	$+x_2$	$+3x_4$	$= 120$
1	x_1	x_1	$+x_2$	$+x_4$	$= 40$
2	x_3		$-x_2$	$+x_3$	$= 20$

Row	BV				RHS
0	x_0	x_0	$+x_3$	$+x_4$	$= 180$
1	x_1	x_1	$+x_3$	$-x_4$	$= 20$
2	x_2	x_2	$-x_3$	$+2x_4$	$= 60$

- Suppose that we add a new nonnegative variable x_5 with the *column* $A^5 = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$ and the objective coefficient $c_5 = 3$.
- Then, we have $\bar{c}_5 := c_{BV}B^{-1}A^5 - c_5 = [1 \quad 1] \begin{bmatrix} 2 \\ 2 \end{bmatrix} - 3 = 1 \geq 0$.
- The current basis is still primal and dual feasible. No need to re-optimize!

Row	BV				RHS
0	x_0	x_0	$+x_3$	$+x_4$	$= 180$
1	x_1	x_1	$+x_3$	$-x_4$	$= 20$
2	x_2	x_2	$-x_3$	$+2x_4$	$= 60$

- Suppose that we add a new nonnegative variable x_5 with the *column* $A^5 = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$ and the objective coefficient $c_5 = 5$.
- Then, we have $\bar{c}_5 := c_{BV}B^{-1}A^5 - c_5 = [1 \quad 1] \begin{bmatrix} 2 \\ 2 \end{bmatrix} - 5 = -1 < 0$ and $B^{-1}A^5 = \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 2 \\ 2 \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \end{bmatrix}$.
- The current basis is not dual feasible! We need to re-optimize.

Modified Tableau:

Row	BV					RHS
0	x_0	x_0	$+x_3$	$+x_4$	$-x_5$	$= 180$
1	x_1	x_1	$+x_3$	$-x_4$		$= 20$
2	x_2	x_2	$-x_3$	$+2x_4$	$+2x_5$	$= 60$

- x_5 enters, x_2 leaves.
- New Tableau (also optimal):

Row	BV					RHS
0	x_0	x_0	$+\frac{1}{2}x_2$	$+\frac{1}{2}x_3$	$+2x_4$	$= 210$
1	x_1	x_1		$+x_3$	$-x_4$	$= 20$
2	x_5		$\frac{1}{2}x_2$	$-\frac{1}{2}x_3$	$+x_4$	$+x_5$ $= 30$

Row	BV				RHS
0	x_0	x_0	$+x_3$	$+x_4$	$= 180$
1	x_1	x_1	$+x_3$	$-x_4$	$= 20$
2	x_2	x_2	$-x_3$	$+2x_4$	$= 60$

- Suppose that we add a new constraint $x_1 + 2x_2 \leq 150$.
- Note that the previous optimal solution satisfies the new constraint as well.
- The current basis is still primal and dual feasible. No need to re-optimize!

Row	BV				RHS
0	x_0	x_0	$+x_3$	$+x_4$	$= 180$
1	x_1	x_1	$+x_3$	$-x_4$	$= 20$
2	x_2	x_2	$-x_3$	$+2x_4$	$= 60$

- Suppose that we add a new constraint $x_1 + 2x_2 \leq 110$.
- Note that the previous optimal solution does NOT satisfy the new constraint.
- The current basis is not primal feasible! We need to re-optimize.

- Modified Tableau:

Row	BV				RHS
0	x_0	x_0	$+x_3$	$+x_4$	$= 180$
1	x_1	x_1	$+x_3$	$-x_4$	$= 20$
2	x_2	x_2	$-x_3$	$+2x_4$	$= 60$
3	x_5	x_1	$+2x_2$	$+x_5$	$= 110$

- Modified Tableau (after the coefficients of the basic variables in the newly added row are zeroed-out):

Row	BV					RHS	
0	x_0	x_0		$+x_3$	$+x_4$	$= 180$	
1	x_1	x_1		$+x_3$	$-x_4$	$= 20$	
2	x_2		x_2	$-x_3$	$+2x_4$	$= 60$	
3	x_5			x_3	$-3x_4$	$+x_5$	$= -30$

Post-Optimality Analysis: Giapetto Example (VI-b)

Row	BV					RHS	
0	x_0	x_0		$+x_3$	$+x_4$	$= 180$	
1	x_1	x_1		$+x_3$	$-x_4$	$= 20$	
2	x_2		x_2	$-x_3$	$+2x_4$	$= 60$	
3	x_5			x_3	$-3x_4$	$+x_5$	$= -30$

- x_5 leaves, x_4 enters.
- New Tableau (also optimal):

Row	BV					RHS
0	x_0	x_0	$+\frac{4}{3}x_3$	$+\frac{1}{3}x_5$	$= 170$	
1	x_1	x_1	$+\frac{2}{3}x_3$	$-\frac{1}{3}x_5$	$= 30$	
2	x_2	x_2	$-\frac{1}{3}x_3$	$+\frac{2}{3}x_5$	$= 40$	
3	x_4		$-\frac{1}{3}x_3$	$+x_4$	$-\frac{1}{3}x_5$	$= 10$

- Obtain the optimal tableau of the Dakota Furniture Company LP:

$$x_0^* = \max 60x_1 + 30x_2 + 20x_3$$

$$\text{s.t. } 8x_1 + 6.5x_2 + 1.5x_3 \leq 48$$

$$4x_1 + 2.5x_2 + 1.5x_3 \leq 20$$

$$2x_1 + 1.5x_2 + 0.5x_3 \leq 8$$

$$x_1, x_2, x_3 \geq 0.$$

- Obtain the optimal tableau using **SimplexTableauWithBV.py**
- Carry out the tasks in HandoutPostoptimality (go to Sucourse), i.e., re-optimize under the following modifications (each part is independent):**
 - Decrease c_1 by 10 units.
 - Increase c_2 by 20 units.
 - Decrease b_1 by 25 units.
 - Increase b_3 by 3 units.
 - Add a new variable with the column $A^7 = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}$ and the objective coefficient $c_7 = 40$.